

Yellow Jacket Swarm Counter Unmanned Aerial System

Project Presentation



Presented by: Parth Patel

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Project High Level Overview

Role:

- Designed and built the interceptor drone entirely. (Ideation, CAD, manufacturing, controls, testing)
- Now, lead 6-person team where I direct the hardware and manufacturing while integrating with SDEs

Timeline

- Last semester: Physical interceptor
- This semester: Drone Detection System
- Aim to reach TRL 6 by April

Program:

- Georgia Tech Create-X Team 7, Interstellar Foundry

Customers:

- DoW, DIU Open Solicitation, Army FUZE Program, SBIR/STTR



The Threat · Why Now

"Peace cannot be kept by force; it can only be achieved by understanding." — Einstein



Low-RCS, Low-Altitude

Commercial drones fly under legacy radar thresholds and Small cross-sections evade S-band sensors designed for aircraft.



Sustainment Cost

Current COTS counter-drone solutions cost \$50K-\$500K per node. Cannot scale to protect dispersed DoD infrastructure.



Fixed-Wing Gap

Fixed-wing interceptors lack agility to track quadcopters making abrupt evasive maneuvers at 25+ mph in close engagements.



OUR SOLUTION

Sub-\$1,000 modular ground station

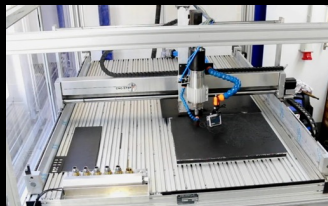
- ✓ NVIDIA Jetson Nano edge computing
- ✓ FMCW radar + CV sensor fusion
- ✓ Kalman filter → eliminates false positives
- ✓ 3× autonomous quadcopter interceptors
- ✓ Waterjet CF frames → < \$280/unit
- ✓ < 15 min deployment on ISV/JLTV

Georgia Tech Create-X · DIU PROJ00656

UAV Interceptor · Manufacturing Approach

HARDWARE SPECS

Frame	450mm quad · waterjet-cut carbon fiber · 3D-printed mounts
Motors	2204 2300KV brushless x 4 per airframe
Props	5" balanced props · optimized for 80+ km/h intercept
Flight Ctrl	Matek F405 running ArduCopter firmware
Comms	ESP32 WiFi MAVLink telemetry to ground station
Cost	~\$280/unit · ~\$1,617 total for 3 swarm C-UAS
Weight	< 10 lb deployable ground station
Deploy	< 15 min setup



DESIGN FOR MANUFACTURABILITY

Waterjet-Cut CF Frames

2D DXF files → waterjet table. Minimal downtime. Consistent tolerance $\pm 0.005"$

3D-Printed Mounts & Arms

PETG/ABS mounts for motors, ESCs, sensors. Rapidly iterate design.

COTS Avionics Stack

Matek F405 + ESP32: proven, in-stock, programmable. No defense foundry lead times. ATO-streamlined.

Cost Scalability

Tier 1: ~\$280/unit MVP.
Tier 2 (6-25): Low-Rate Production.
Tier 3 (26-100+): mass manufacturing.

Flight Physics · Design Rationale

PROPULSION PHYSICS

Hover condition:

$$4T = mg \quad | \quad T = (1/2)\rho A v_i^2$$

$T_{total} \sim 120 \text{ N}$, $mass = 0.280 \text{ kg} \rightarrow T/W = 43.7$

Drag at 80 km/h (22 m/s):

$$F_{drag} = (1/2) \rho C_d A v^2$$

$$= (1/2) (1.2) (0.4) (0.04) (22)^2 = 4.7 \text{ N}$$

Kinetic impact energy at intercept:

$$KE = (1/2) m v_{rel}^2$$

Head-on closing @ 44 m/s: $KE \sim 271 \text{ J}$

Stress in CF frame (motor mount):

$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2}$$

FEA: 48 MPa at mount holes, $FS > 12$

DESIGN CHOICES

450mm frame selection

Longer moment arm $\rightarrow$ torque per RPM gain. Larger props $\rightarrow$ efficiency. Tradeoff: agility vs. 250mm race frame. Intercept geometry prioritizes speed over maneuverability.

2204 2300KV motor selection

High KV $\rightarrow$ high RPM at 3S $\rightarrow$ fast prop speed $\rightarrow$ agility. $T/W > 4$ at hover. $> 75\%$ thrust reserve for intercept acceleration from 0 to intercept $< 3 \text{ sec}$.

3K twill CF plate (3mm)

$\sigma_{UTS} \sim 600 \text{ MPa}$. Waterjet-cuttable with $\pm 0.005''$ tolerance. Consistent cross-section vs. hand-layup. Safety factor > 12 at motor mounts under crash load.

Polycarbonate guards over CF-PLA

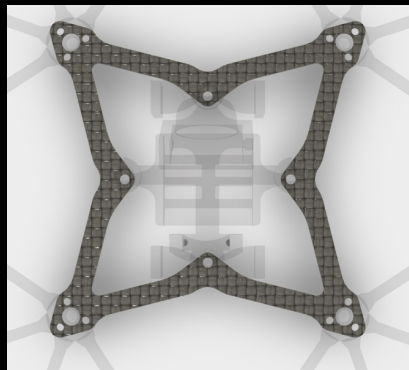
CF-PLA: 220N peak load but catastrophic brittle fracture. PC: 150N but plastic deformation. Failure MODE trumps failure LOAD for attritable interceptors. Must survive, not just endure.

Interceptor Design · CAD Modeling

Airframe: 450mm quadcopter configuration. Waterjet-cut carbon fiber main plate (3K twill weave, 3mm thickness). CAD modeled in Fusion 360. 3D-printed PETG motor mounts and camera bracket assembly.

Propulsion: 2204 2300KV brushless motors × 4, 5" balanced propellers, optimized for 80+ km/h intercept.

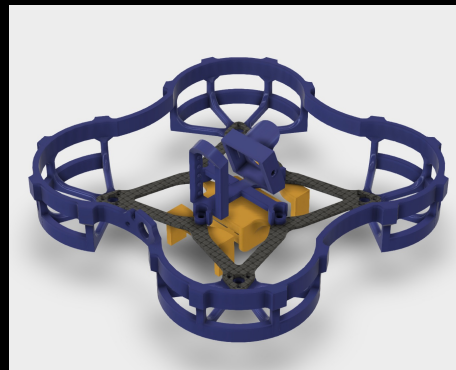
Flight Control: Matek F405 running ArduCopter 4.3. ESP32 MAVLink telemetry bridge. Coordinated engagement vectors uplinked from ground station via Python API.



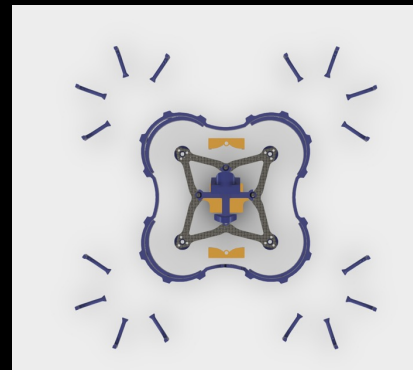
CF Frame (top view)
Waterjet DXF → waterjet table



Payload + Camera Mount
3D-printed PETG



Full Assembly: CF frame (gray) +
guards (blue) + mounts (gold)



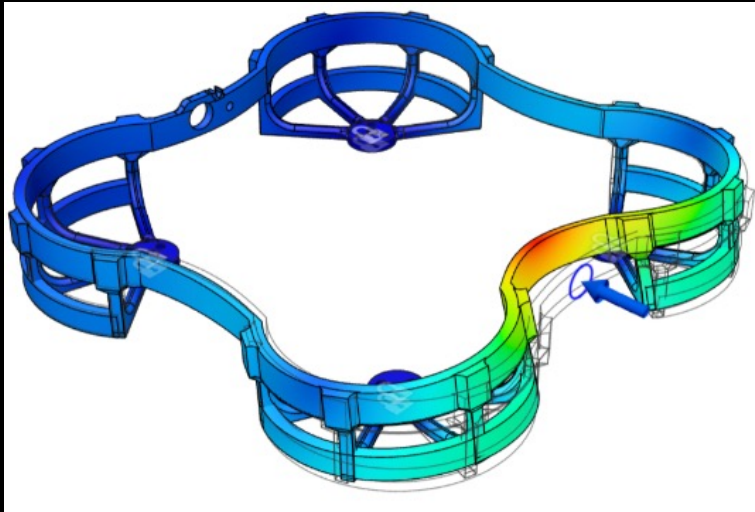
Full Assembly Exploded View

Material Selection · Structural FEA

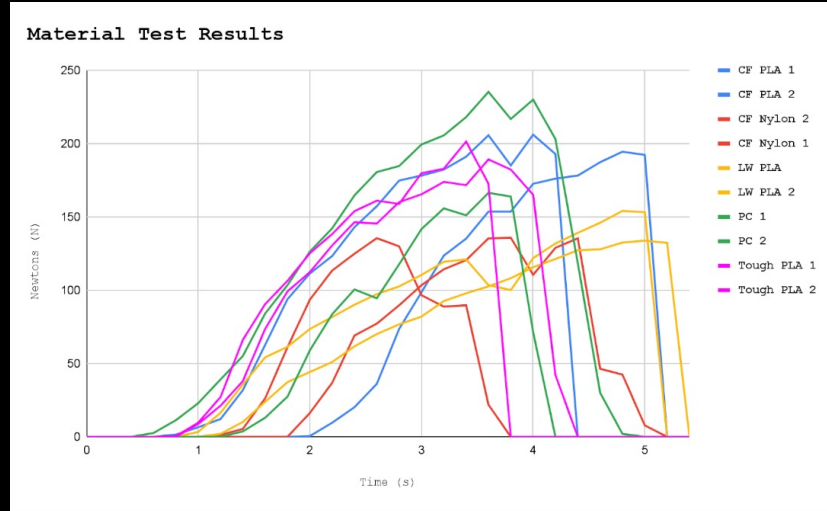
Test Campaign: 10 prop guard specimens across 5 filament types (PETG-CF, CF-PLA, CF-Nylon, LW-PLA, Polycarbonate) loaded to failure under axial compression. Peak load plotted over time to characterize both maximum strength and failure mode (brittle fracture vs. progressive deformation).

Key Finding: CF-PLA achieved peak loads exceeding 220N but fractured catastrophically. Polycarbonate showed the best post-failure behavior (plastic deformation rather than shattering), selected as primary material for guards.

$$\text{Von Mises Stress: } \sigma_{vm} = \sqrt{(\sigma_{\theta}^2 + \sigma_{ax}^2 - \sigma_{\theta} \cdot \sigma_{ax})}$$



ANSYS FEA: Prop guard structural analysis
Von Mises stress distribution under axial compression load.



Material test result: Force (N) vs. time (s) across 10 specimens.

Physical Build Setup

Carbon Fiber Main Plate: 2D DXF exported from Fusion 360 and cut on a waterjet table ($\pm 0.005''$ tolerance).

- **Challenge:** delamination with carbon fiber (piercing in scrap material, using lower pressure)

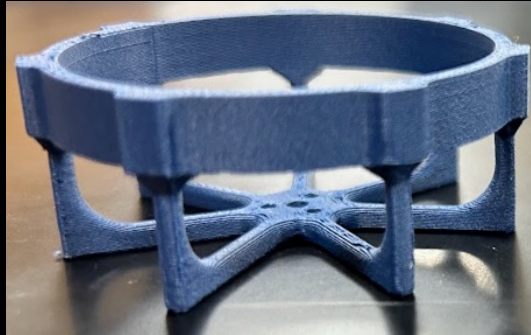
3D-Printed Components: Motor mounts, prop guards, and camera brackets printed in PETG and polycarbonate using Bambu Lab X1 Carbon. Iterated across 5 revs to achieve target budget and impact resilience.

- **Challenge:** Difficulty with polycarbonate (requires high enclosure temps to prevent warping)

Assembly Budget: ~\$280/interceptor. Total 3-unit CUAS: ~\$1,617.



Waterjet CF frame
(physical part)



3D-printed motor guards



3D-printed prop guards



Camera/Payload mount
finished part

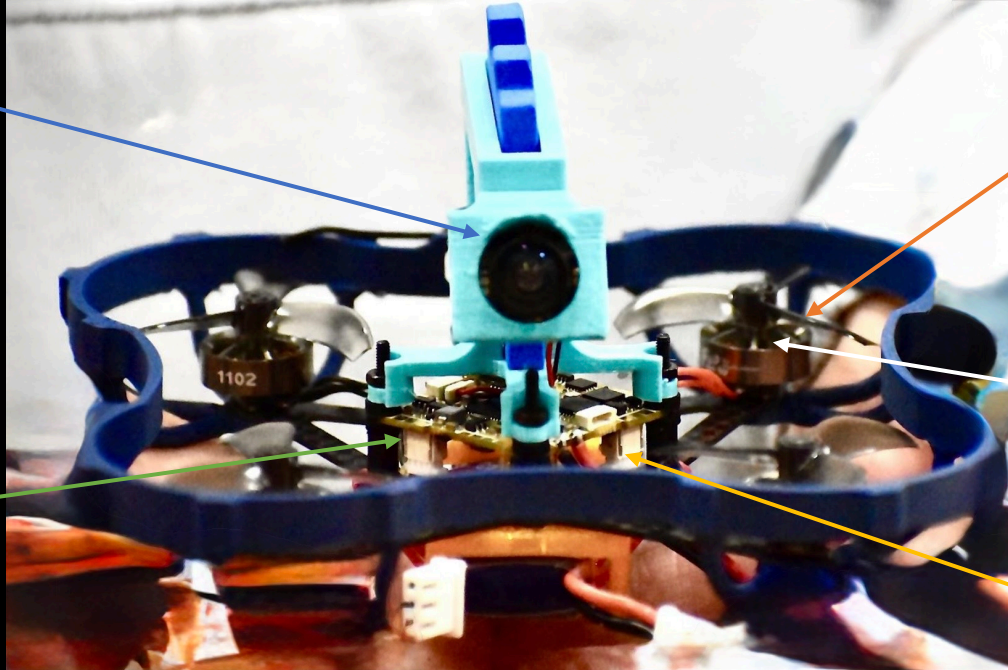
Final Assembly · COTS Parts



Bee Eye



BLV4



Flow 1102

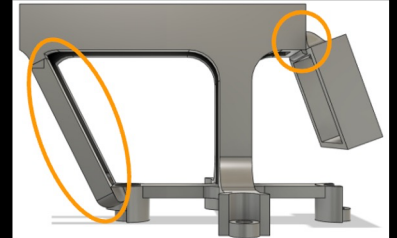
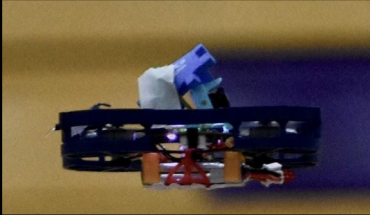
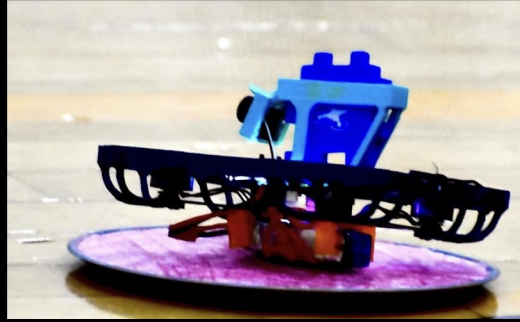


Azi Triblade



XT-30

Initial Flight Testing



Upon Testing,
realized the camera
mount was not strong
enough, so decided
to add structural
supports to the
camera mount itself.

Failures · Adjustments



Camera mount collapsed in flight

Vibration + thrust reversal loads not in single post design.



PID oscillations / unstable hover

ArduCopter defaults tuned for large/stiff frames; sub-300g frame has different inertia tensor.



Prop guard CF-PLA: brittle fracture in crash

CF-PLA 220N peak but catastrophic shatter; debris hazard, system unflyable.



Camera FOV missed target in terminal intercept phase

Level-mounted camera; approach from above geometry requires forward down FOV.



3-point triangulated PETG mount, silicone damping grommets.

Design for dynamic loads, not static geometry. Verify with first principles



Tuned PIDs from scratch; notch filter at 200Hz; Blackbox logging.

Never assume firmware defaults work. Tune methodically.



Tested 5 materials; selected PC (150N, plastic deformation, re-flyable).

Failure MODE > peak strength.

Ductile failure > Tensile Strength



-15 deg angled mount; recalibrated YOLO bounding box coordinates

Model engagement geometry in 3D first. Kinematic analysis before hardware.

Detection System Architecture

01



Primary Detection

RADAR

24 GHz FMCW
K-band Doppler
Range + velocity
≤20 dBm emission

02



Secondary Tracking

OPTICAL

OAK-D Pro stereo
Hikvision 8MP
Angular precision
Object classification

03



Edge Compute

JETSON

NVIDIA Jetson Nano
60+ TOPS, 7-25W
YOLOv8 inference
No cloud dependency

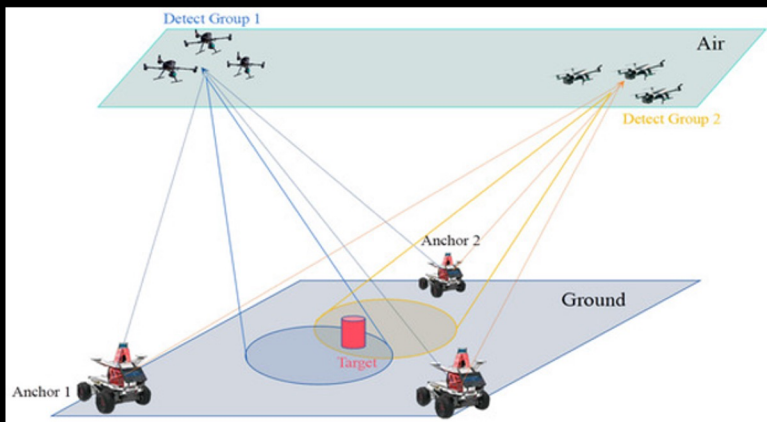
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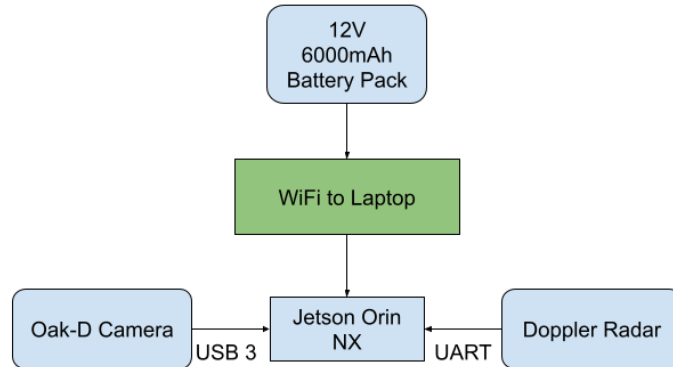
Sensor Fusion

HANDSHAKE

Kalman filter
Fuses radar + optics
Elim. false positives
Locks Group 1/2 sUAS



Field-tested Casing



Sensor Fusion Physics · Kalman Filter

The math that locks Group 1/2 targets while eliminating birds and weather clutter

RADAR DETECTION

Doppler frequency shift:

$$f_D = 2v_r \cdot f_c / c$$

Radar range equation:

$$R = [(P_t \cdot G_t \cdot G_r \cdot \lambda^2 \cdot \sigma) / ((4\pi)^3 \cdot S_{min})]^{(1/4)}$$

FMCW range resolution:

$$\Delta R = c / (2 \cdot B) \quad \text{where } B = \text{bandwidth}$$

Velocity resolution:

$$\Delta v = \lambda / (2 \cdot T_{CPI})$$

f_D : Doppler frequency shift, v_r : Relative velocity, f_c : Carrier frequency,
 c : Speed of light, R : Target range, P_t : Transmit power, G_t : Transmit antenna gain,
 G_r : Receive antenna gain, λ : Wavelength, σ : Radar cross section,
 S_{min} : Minimum detectable signal, ΔR : Range resolution, B : Bandwidth,
 Δv : Velocity resolution, T_{CPI} : Coherent processing interval

360°

Azimuth Coverage

10 Hz

Update Rate

KALMAN FILTER FUSION

State prediction:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F} \cdot \hat{\mathbf{x}}_{k-1} + \mathbf{B} \cdot \mathbf{u}_k$$

Covariance prediction:

$$\mathbf{P}_{k|k-1} = \mathbf{F} \cdot \mathbf{P}_{k-1} \cdot \mathbf{F}^T + \mathbf{Q}$$

Kalman gain:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \cdot \mathbf{H}^T \cdot (\mathbf{H} \cdot \mathbf{P}_{k|k-1} \cdot \mathbf{H}^T + \mathbf{R})^{-1}$$

State update:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \cdot (\mathbf{z}_k - \mathbf{H} \cdot \hat{\mathbf{x}}_{k|k-1})$$

$\hat{\mathbf{x}}_{k|k-1}$: Predicted state estimate, $\mathbf{F}$: State transition model, $\mathbf{z}_k$: Measurement vector
 $\hat{\mathbf{x}}_{k-1}$: Previous state estimate, $\mathbf{B}$: Control input model, $\mathbf{u}_k$: Control vector,
 $\mathbf{P}_{k|k-1}$: Predicted covariance estimate, $\mathbf{Q}$: Process noise covariance, $\mathbf{K}_k$: Kalman gain,
 $\mathbf{H}$: Observation model, $\mathbf{R}$: Measurement noise covariance, $\hat{\mathbf{x}}_k$: Updated state estimate

<5 mrad

Az/El Error

50 mph

Vehicle Track Speed

Hardware Subsystems

Radar (Tiered Scaling) —>

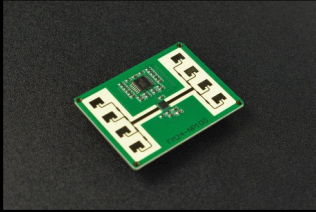
Optical Suite —>

Edge Processing —>

Tactical Footprint



mmWave Microwave Radar
(~\$66) for MVP and >30m tracking.



Highly compatible with
OAK-D Pro (RGB/stereo depth) or



NVIDIA Jetson Nano delivers localized, cloud-independent AI (60+ TOPS) on a highly efficient **7-25W power draw.**



Sub-10 lb modular design supports under 15-minute deployment on ISV, JLTV, FMTV, and HEMTT platforms.



Software Subsystems

GNSS-Denied Navigation —>

Spectral Stealth —>

Cybersecurity



Converts 2D pixels to 3D map coordinates using OpenCV and SLAM to localize threats without GPS.



Air-gapped serial communication and passive optics ensure maximum EMI robustness and low physical/spectral signatures.



Architecture strictly complies with **DOD Instruction 8510.01 (RMF)** to streamline the Authorization to Operate (ATO).

LOE 1 Homeland

ASCII output · FAAD C2 · 5-30kW gen compatible

LOE 2 Tactical/Mobile

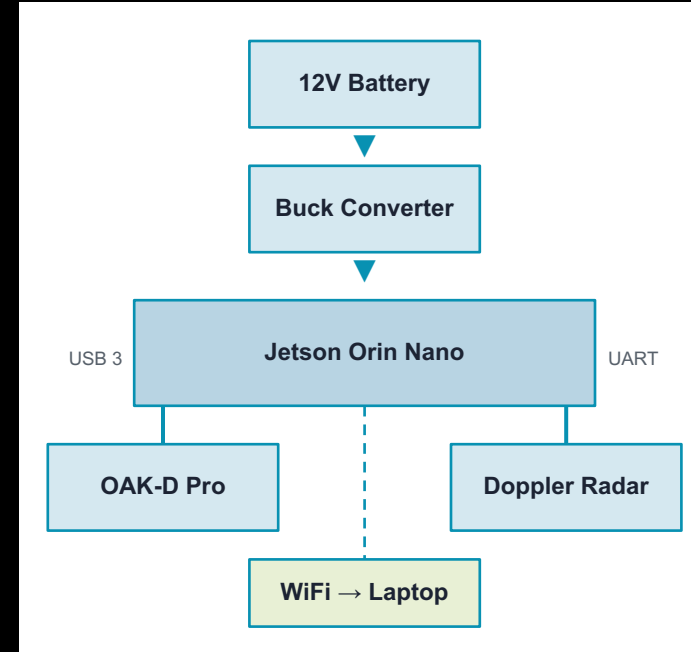
GNSS-denied · ISV/JLTV mount · passive posture

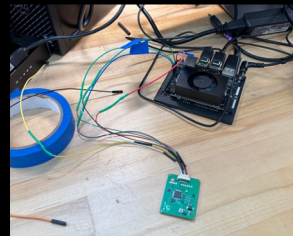
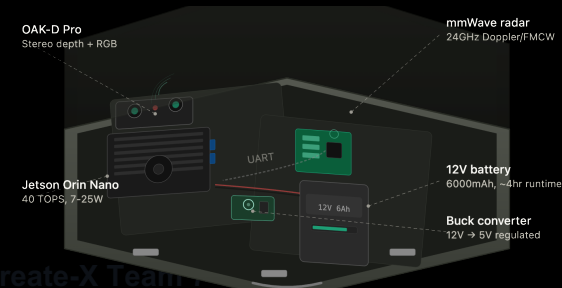
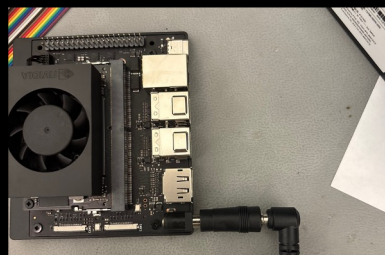
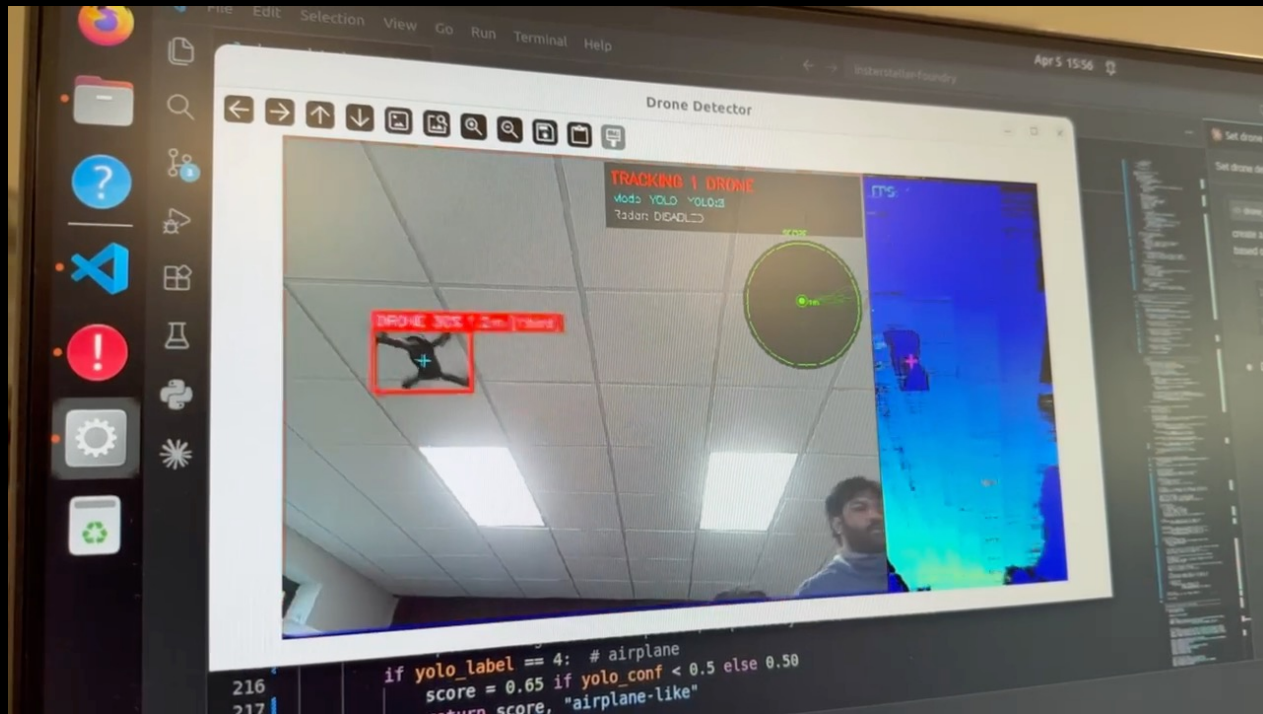
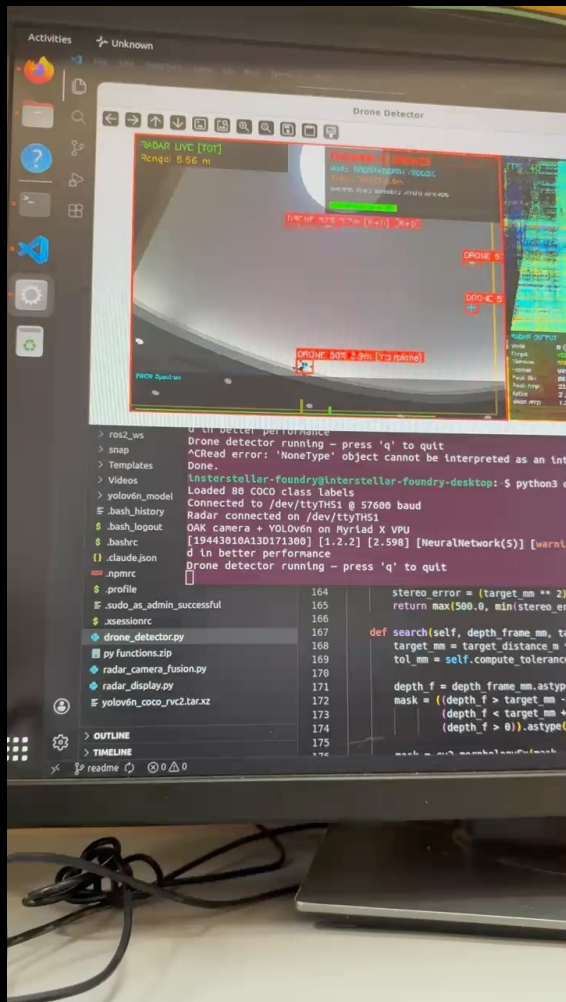
BILL OF MATERIALS

Component	Supplier	Cost
OAK-D Pro Camera	Neobits	\$302.19
Jetson Orin Nano	Amazon	\$245.00
12V 6000mAh Battery	Fat Tire House	\$65.00
mmWave 24GHz Radar	DFRobot	\$65.90
DC Converter + Cables	Amazon	\$27.07
Test Drones	Amazon	\$39.98
Sandisk MicroSD	Amazon	\$56.40
Tax + AI Tools	Various	\$175
TOTAL		\$957.15

Left to purchase: Pelican Case <\$100

Hardware Schematic





Engagement Logic · Swarm Software

GROUND STATION SOFTWARE

Python / OpenCV

Real-time video processing pipeline

HB100 Doppler

Velocity + range primary detection

Kalman (FilterPy)

Multi-target tracking + fusion

YOLOv8 inference

Jetson-accelerated classification

PyQt6 UI

Operator display / engagement cue

ENGAGEMENT LOGIC

Target acquired

Dual-sensor confirm → lock

Primary dispatch

Lead interceptor → head-on vector

Flankers launch

2 flankers on diverging intercept

Abort logic

Flankers abort if primary scores
kill

Re-engage

Missed intercept → re-cue loop

MVP SUCCESS CRITERIA

Dual-sensor detect

Both radar AND camera confirm

80+ km/h targets

Intercept at full speed

3 simultaneous launches

Coordinated swarm execution

Kinetic contact

Physical intercept confirmed

< 15 min deploy

From packed to combat ready

Business Case · Current Market

The gap: low-cost, high-lethal, mass-producible C-UAS at scale

TIERED ROM PRICING

TIER 1 1-5 units

\$500-\$1,000/node

Prototype / evaluation

TIER 2 6-25 units

LRIP pricing

Low-rate initial production

TIER 3 26-100+ units

Mass manufacturing

Full contract scale

MARKET CONTEXT

\$4B+

C-UAS global market by 2030

~\$500K

Avg. legacy sensor node cost

<\$1K

Our node cost is 500× cheaper

Group 1/2

Primary threat is underserved

Thank you!

Parth Patel

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